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Introduction

This document presents the design, development, and analytical outcomes of an interactive CEO dashboard built using Microsoft Power BI. Power BI was selected as the development platform because it is the industry-leading business intelligence solution for creating interactive, enterprise-grade dashboards [3]. All data preparation, modelling, DAX programming, and report design were completed in Power BI Desktop. The finished report was then published to Power BI Service, Microsoft's cloud platform [3], making it accessible across desktop, tablet, and mobile devices with responsive layout scaling.

The dashboard is built for the CEO to make informed business decisions, providing the financial metrics and geographic insights needed to act with clarity and confidence. A drill-through page is also created to help the CEO develop strategic plans and organise targeted campaigns to increase sales and profit. The report spans two analytical pages and one drill-through page, all tied together by a synchronised Year-Month slicer and the CIC brand theme.

BI Development Cycle

1. Data Gathering

The dataset used is the Superstore Dataset, a retail simulation dataset sourced from Kaggle [4]. It contains 9,994 order records spanning product categories, customer segments, geographical dimensions, and business metrics including sales, profit, discount, and quantity. These records cover four years (2014–2017) across the United States.

For this project, the dataset was rebranded as "Chetan in Cork" (CIC) - a fictional retail company identity created to provide a purposeful business context and a consistent visual theme. This renaming was applied at the table level in Power Query so that all DAX measures and visual field references consistently use the CIC table name.

2. Data Understanding

The Superstore dataset contains multiple dimension groups:

- Order dimensions: Row ID, Order ID, Order Date, Ship Date, Ship Mode capturing when orders were placed, their shipment duration, and mode of delivery.

- Customer dimensions: Customer ID, Customer Name, Segment identifying buyers across Consumer, Corporate, and Home Office segments.
- Geographic dimensions: Country, City, State, Postal Code, and Region enabling spatial analysis at city, state, and regional level across the West, East, Central, and South US regions.
- Product dimensions: A three level hierarchy of Product ID, Category, Sub-Category, and Product Name spanning Furniture, Office Supplies, and Technology across 17 subcategories.
- Financial dimensions: Sales (total revenue), Quantity (units sold), Profit, and Discount (order-level rate applied) the primary performance indicators.

The dashboard is built to help the CEO make informed business decisions by surfacing financial and geographic insights needed to act with clarity. A drill-through page assists the CEO in creating strategic plans and organising targeted campaigns to grow sales and profit.

3. Data Validation

Raw data was profiled in Power Query using Column Quality, Column Distribution, and Column Profile views [2]. This step is critical because it highlights error values, unexpected data types, outliers, and format inconsistencies before modelling begins.

The primary issue identified was in the Order Date column, which Power Query detected as text rather than a date data type. On inspection, two distinct date formats were mixed within the same column some records used DD/MM/YYYY and others used MM/DD/YYYY. All remaining columns passed validation with consistent formatting and correct data types.

4. Data Wrangling

Data wrangling was performed in Power Query, Power BI's transformation engine, with all steps recorded as M code in the Applied Steps pane [2]. The mixed date format was resolved using a custom M expression that inspects each date string and selects the correct locale-specific parser, The logic is if the leading two-digit number exceeds 12, it cannot represent a month, so the en-GB (DD/MM/YYYY) locale is applied; if 12 or below, en-US (MM/DD/YYYY) is used. Once the clean Date column was created, it was split into Day, Month, and Year integer columns using Table.SplitColumn. Correct data types were applied to all columns and rows were sorted ascending by Date. A separate Date dimension table was then created and a one-to-many relationship established from the Date Table to the CIC dataset in the modelling pane, forming the required star schema.

5. DAX Development

DAX (Data Analysis Expressions) is the formula language built into Power BI for creating calculated columns, calculated tables, and measures [1]. Nine measures were developed to support the dashboard visuals.

Base Aggregations:

Sales = SUM(CIC[Sales])

Profit = SUM(CIC[Profit])

Product Cost = SUM(CIC[Sales]) - SUM(CIC[Profit])

As the dataset contains no direct cost column, Product Cost is derived by subtracting Profit from Sales.

Ratio Measures:

Profit Percentage = DIVIDE([Profit], [Sales])

Avg Discount % = AVERAGE(CIC[Discount])

DIVIDE is used in place of the division operator to return blank when Sales is zero, avoiding divide-by-zero errors [1].

Time-Intelligence Measures:

CY_Profit = CALCULATE([Profit], 'Date Table'[Date])

LY_Profit = CALCULATE([Profit], SAMEPERIODLASTYEAR('Date Table'[Date]))

YOY Growth = DIVIDE([Profit] - [LY_Profit], [LY_Profit])

CALCULATE is the most powerful DAX function, modifying filter context before evaluating an expression [1]. SAMEPERIODLASTYEAR shifts the date filter back exactly 12 months, requiring a continuous Date Table - the reason the star schema is essential. YOY Growth involves three-level nesting: it references Profit and LY_Profit; LY_Profit nests Profit inside CALCULATE with SAMEPERIODLASTYEAR; Profit aggregates CIC[Profit] through the Date Table relationship. This nesting modifies filter context twice before reaching the base aggregation.

Dashboard Development

1. Sales and Profit Overview

Six KPI cards provide an immediate financial snapshot of Total Sales, Total Profit, Profit Margin %, Total Orders, Average Discount %, and YoY Growth. Below these, a dual-line monthly trend chart shows

how Sales and Profit have moved together over time. A Waterfall chart decomposes Profit by category, making gains and losses immediately visible. A Donut chart displays revenue by customer segment with Profit surfaced in the tooltip. The Key Influencers AI visual rounds out the page by identifying which variables statistically drive Profit up or down.

2. Geographic and Regional Analysis

A filled map with city level Sales bubbles provides spatial context for performance. A scatter plot of Sales versus Profit by Region and Category reveals margin efficiency across the portfolio. A Ribbon chart shows how regional Profit rankings have shifted year on year. A Sales by Region bar chart serves as the drill-through trigger right clicking any region bar navigates to the Campaign Planning page with that region pre-filtered.

3. Drill-Through Page - Campaign Planning

The drill-through page opens with filter context applied to whichever region was right-clicked on the Page 2 bar chart. Four slicers (Region, City, Category, Sub-Category) allow further narrowing of the data. A matrix table shows Region, City, Sales, Profit, Average Discount %, Profit %, YoY Growth, and Count of Customers, with conditional formatting applied to the Sales and Profit columns to highlight underperforming cities immediately.

4. Publishing to Power BI Service

The report was published to Power BI Service - Microsoft's cloud platform for sharing and managing Power BI content [3]. Publishing uploads the .pbix file to a designated workspace in the cloud, after which the report is accessible via a web browser or the Power BI mobile app without requiring Power BI Desktop to be installed.

Q&A

a) Where Did the Idea Come From?

The idea and concept for this dashboard originated from professional work experience. While working as a Data Consultant, a similar CEO-level dashboard was designed and developed for AstraZeneca, a global pharmaceutical company. That project involved building executive-facing Power BI reports with KPI tracking, regional performance analysis, and drill-through pages for operational planning a workflow that directly inspired the structure of this CIC dashboard.

b) Has Anyone Else Visualised This Dataset?

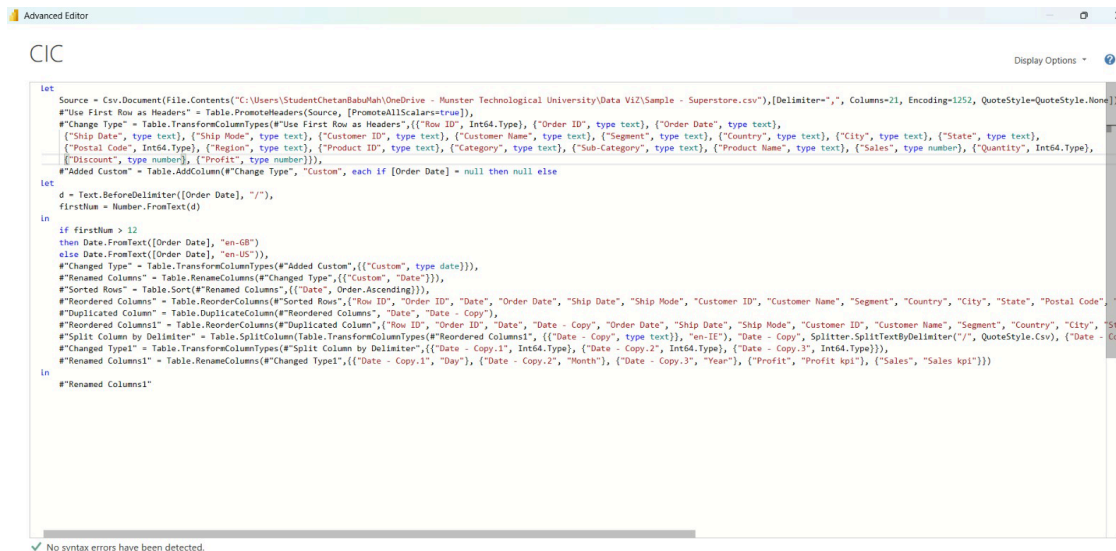
Yes, the Superstore dataset has been widely used in the data visualization community, and several dashboards have been published online. One notable example is the Superstore dashboard showcased by Tableau, which demonstrates how the dataset can be analyzed through interactive dashboards focusing on business performance indicators such as sales, profit ratio, and shipping metrics [5]. While this dashboard provides valuable analytical insights, the approach taken in this assignment differs in its design objective and narrative structure. The dashboard developed for this assignment was designed as a CEO-level decision-support dashboard, focusing on strategic performance monitoring rather than exploratory analysis alone.

c) What Is Novel About This Approach?

Most published visualizations of this dataset are built for data analysis. This dashboard was built for CEO need to make a fast, informed decision about where to grow sales and profit. Every visual and measures was built to support the CEO decision making. The result is a dashboard where a CEO can open the report, read the KPI cards in seconds, spot the South region trailing the West by approximately \$400,000, drill through to the campaign planning page, and walk out of a board meeting with a clear action all without needing to understand the data from analyst. This quick informed business decisions are what distinguishes from any work published for this dataset.

d) Complex Implementation Detail

The complex part was to handle and creating a separate date dimension which is handled with a M query. This is the m query used for creating a custom date column and date table.



```
let
    Source = Csv.Document(File.Contents("C:\Users\Student\ChetanBabu\OneDrive - Munster Technological University\Data Viz\Sample - Superstore.csv"),[Delimiter=";", Columns=21, Encoding=1252, QuoteStyle=QuoteStyle.None]),
    #"Use First Row as Headers" = Table.PromoteHeaders(Source, [PromoteAllScalars=true]),
    #"Change Type" = Table.TransformColumnTypes(#"Use First Row as Headers",{{"Row ID", Int64.Type}, {"Order ID", type text}, {"Order Date", type text}, {"Ship Date", type text}, {"Ship Mode", type text}, {"Customer ID", type text}, {"Customer Name", type text}, {"Segment", type text}, {"Country", type text}, {"City", type text}, {"State", type text}, {"Postal Code", Int64.Type}, {"Region", type text}, {"Product ID", type text}, {"Category", type text}, {"Sub-Category", type text}, {"Product Name", type text}, {"Sales", type number}, {"Quantity", Int64.Type}, {"Discount", type number}, {"Profit", type number}}}),
    #"Added Custom" = Table.AddColumn(#"Change Type", "Custom", each if [Order Date] = null then null else
let
    d = Text.BeforeDelimiter([Order Date], "/"),
    firstNum = Number.FromText(d)
in
    if firstNum > 12
    then Date.FromText([Order Date], "en-GB")
    else Date.FromText([Order Date], "en-US"),
    #"Changed Type" = Table.TransformColumnTypes(#"Added Custom",{{"Custom", type date}}),
    #"Renamed Columns" = Table.RenameColumns(#"Changed Type",{{"Custom", "Date"}}),
    #"Sorted Rows" = Table.Sort(#"Renamed Columns",{{"Date", Order.Ascending}}),
    #"Reordered Columns" = Table.ReorderColumns(#"Sorted Rows",{"Row ID", "Order ID", "Date", "Order Date", "Ship Date", "Ship Mode", "Customer ID", "Customer Name", "Segment", "Country", "City", "State", "Postal Code", "Sales", "Quantity", "Discount", "Profit"}),
    #"Duplicated Column" = Table.DuplicateColumn(#"Reordered Columns", "Date", "Date - Copy"),
    #"Reordered Columns1" = Table.ReorderColumns(#"Duplicated Column",{"Row ID", "Order ID", "Date", "Date - Copy", "Order Date", "Ship Date", "Ship Mode", "Customer ID", "Customer Name", "Segment", "Country", "City", "State", "Postal Code", "Sales", "Quantity", "Discount", "Profit"}),
    #"Split Column by Delimiter" = Table.SplitColumn(Table.TransformColumnTypes(#"Reordered Columns1", {{"Date - Copy", type text}, {"en-IE", "Date - Copy", Splitter.SplitTextByDelimiter("/", QuoteStyle.Csv), {"Date - Copy1", Int64.Type}, {"Date - Copy2", Int64.Type}, {"Date - Copy3", Int64.Type}}}), "Date - Copy", Splitter.SplitTextByDelimiter("/", QuoteStyle.Csv), {"Date - Copy1", Int64.Type}, {"Date - Copy2", Int64.Type}, {"Date - Copy3", Int64.Type}),
    #"Renamed Columns1" = Table.RenameColumns(#"Split Column by Delimiter",{{"Date - Copy1", "Day"}, {"Date - Copy2", "Month"}, {"Date - Copy3", "Year"}, {"Profit", "Profit kpi"}, {"Sales", "Sales kpi"}}})
in
    #"Renamed Columns1"
```

✓ No syntax errors have been detected.

e) Three Design Considerations

i. Company Theme and Stakeholder Connection -

Applying a branded visual theme to a dashboard matching colours, fonts, and logo identity to the company is standard practice in professional BI development. When a CEO or regional director opens a report that uses their company colours and branding, it creates an immediate sense of ownership and trust in the data. For CIC, the green-and-white palette taken from the logo was applied to all headings, KPI cards, and section dividers. This personalisation is widely used in industry precisely because it bridges the gap between a technical data product and a business communication tool stakeholders engage more readily with something that looks and feels like it belongs to their organisation.

ii. Storytelling and Visual Placement -

Eye Movement Design. The layout of each dashboard page was designed around the natural eye movement pattern of a reader, which follows a path from the top-left corner across and downward. On Page 1, the six KPI cards are placed at the top the first thing the eye lands on delivering the sales numbers immediately. The left-to-right, simple-to-complex placement is a standard storytelling practice that guides the CEO through the data narrative without leaving them to find their own path.

iii. Audience and attention to detail-

One of the foundational principles of business intelligence is that no dashboard can be effective unless the developer truly understands both the data and the people who will use it. In this project, the data understanding phase mapped all five dimension groups - Order, Customer, Geographic, Product, and Financial. The audience understanding shaped the three-page structure: the CEO gets the financial summary, the analyst gets the geography, and the regional manager gets the drill-through operational detail. This approach ensures that no stakeholder requirement is missed and that every visual serves a specific decision-making need.

f) What Aspect of Data Analysis Was Investigated?

The dashboard covers three types of analysis.

Descriptive analysis is handled by the KPI cards and the monthly trend chart, giving a clear picture of where the sales numbers.

Distribution comparison appears in the donut chart and the regional bar chart, showing how revenue is spread across segments and geographies.

Statistical/AI-driven analysis is delivered through the Key Influencers AI visual, which identifies which categorical and numerical variables statistically increase or decrease Profit.

g) How Could the Work Be Improved?

First, the dataset includes a Ship Mode dimension which consists of Standard Class, Second Class, First Class, and Same Day which is currently only surfaced as a tooltip on the map visual. With more time, a dedicated Shipment Mode analysis page would be developed, examining how fulfilment method affects order volume, delivery performance, and profitability across regions and categories. For a CEO thinking about supply chain efficiency, understanding whether Same Day shipping generates sufficient margin premium to justify its cost is exactly the kind of operational insight a CEO dashboard should answer.

Second, the South region sales gap of approximately \$400,000 behind the West deserves a more focused investigation. The drill-through page provides a starting point, but a dedicated South analysis could include customer acquisition trends over time, sub-category performance benchmarked against the West, and discount rate analysis by city - giving

leadership a complete picture of whether the gap is structural or addressable through targeted commercial action.

References

- [1] Microsoft, "DAX function reference," Microsoft Learn. [Online]. Available:<https://learn.microsoft.com/en-us/dax> [Accessed: Mar. 2025].
- [2] Microsoft, "Power Query M formula language reference," Microsoft Learn. [Online]. Available:<https://learn.microsoft.com/en-us/powerquery-m> [Accessed: Mar. 2025].
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- [4] V. Vivek, "Superstore Dataset," Kaggle. [Online]. Available:<https://www.kaggle.com/datasets/vivek468/superstore-dataset-final> [Accessed: Mar. 2025].
- [5] Tableau, "Superstore Dashboard Showcase," Tableau Public. [Online]. Available:<https://www.tableau.com/data-insights/dashboard-showcase/superstore> [Accessed: Mar. 2025].